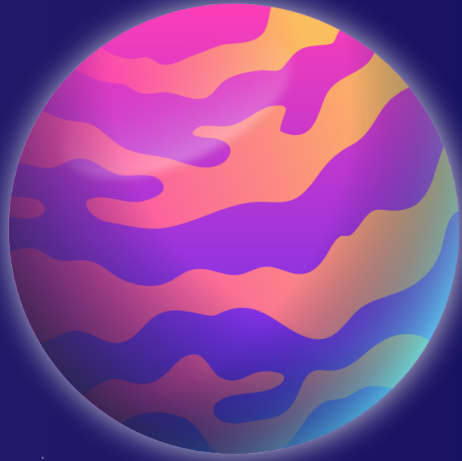


How to write a thesis?

By Dr. Sambatriniaina Rajohnson





What is one thing that worries
you about writing a thesis?



Anatomy of a Thesis (IMRED)*

1. Abstract
2. Introduction (Literature review)
3. Methods
4. Results
5. Discussion
6. Conclusion
7. Bibliography
8. Appendices

* Please refer to your university requirements for more detailed requirements for formatting (e.g., number of pages, page margins, etc.)

Group Activity (30 mn)

Group 1 (Abstract): Kantoarivelo, Matoky

Group 2 (Intro): Miora, Arnaulvine, Francia

Group 3 (Methods): Nancy, Nomena, Gabryella

Group 4 (Results): Isaia, Kantoniavo, Olivier

Group 5 (Discussion): Dera, Fameno, Avotra

Group 6 (Conclusion): Ranto, Herman

Group 7 (Biblio): Ny Antsaly, Mamy

Group 8 (Append.): Laza, Anjia

Discuss in 10-12 mn:

What goes in this section?

What is its purpose?

What mistakes do you think are often made?

What tools or strategies can help to get a good section?

Abstract (MSc: half to 1 page, PhD: 1-2 pages)

- 01 **What goes in:** Summary of your whole thesis (problem, methods, main findings, conclusion).
- 02 **Purpose:** Help readers quickly grasp what the thesis is about and why it matters.
- 03 **Common mistakes:** Too vague, too long, missing key findings or conclusions.
- 04 **Strategy:** Write it last when your thesis is complete.
- 05 **Tool:** Use a 1-sentence structure: **Background – Problem – Method – Results – Impact.**

Introduction & Literature review

- 01 **What goes in:** Scientific background, context, open questions, limitations, and your thesis goals.
- 02 **Purpose:** Show what has been done, what is missing, and **why your work matters.**
- 03 **Common mistakes:** Writing a history of the field instead of building a rationale. You need to show how your research fits in!
- 04 **Strategy:** Structure as a funnel (**broad to narrow**), ending in your research question.
- 05 **Tool:** Use a citation manager to organize references (see later slides)

Methods

- 01 **What goes in:** How you did your research – observations, tools, code, data, analysis steps.
- 02 **Purpose:** Allow others to **replicate** or **validate** your results.
- 03 **Common mistakes:** Being too brief or skipping assumptions and calibration details.
- 04 **Strategy:** Be clear and systematic – think of it as a recipe.
- 05 **Tool:** Use diagrams, flowcharts, tables, for clarity.

Results

- 01 **What goes in:** Your actual findings — plots, maps, statistics, detections, measurements.
- 02 **Purpose:** Present evidence without interpretation.
- 03 **Common mistakes:** Mixing results and discussion, or overloading with raw plots. Choose either a plot or a table!
- 04 **Strategy:** Use subtitles per result theme; include uncertainties.
- 05 **Tool:** Use Python/LaTeX plotting tools and caption all figures clearly.

Discussion

- 01 **What goes in:** Interpretation of the results, comparison with other work, implications.
- 02 **Purpose:** Explain **what your results mean** and how they fit the big picture.
- 03 **Common mistakes:** Repeating results or being speculative without support.
- 04 **Strategy:** Address each research question in the intro; mention limitations and future work.
- 05 **Tool:** Read papers for comparison. Sometimes, people can also combine results & discussion together.

Conclusion

- 01 **What goes in:** Summary of main findings, their significance, and future directions.
- 02 **Purpose:** Give closure and highlight your contribution.
- 03 **Common mistakes:** Introducing new material or being too vague. Just repeating the results without synthesizing.
- 04 **Strategy:** Revisit your objectives and clearly state how they were achieved.
- 05 **Tool:** Keep it short (usually 1–2 pages max); bullet format is OK.

Bibliography

- 01 **What goes in:** All cited works in your thesis.
- 02 **Purpose:** Acknowledge previous research, provide traceability.
- 03 **Common mistakes:** Inconsistent format or missing entries. Every website reference must include the **date of access** since websites can change.
- 04 **Strategy:** Update as you go, don't leave it for the end.
- 05 **Tool:** Use BibTeX (LaTeX) or citation software to auto-format. For minor mentions such as websites, one can use footnotes.

Appendices

- 01 **What goes in:** Supplementary materials like code, tables, extra plots, calibration.
- 02 **Purpose:** Support your thesis without interrupting the main flow.
- 03 **Common mistakes:** Hiding essential information that should be in the Methods, you need to identify if it is a core detail or not.
- 04 **Strategy:** Reference them in the main text when needed.
- 05 **Tool:** Use clear titles (“Appendix A: Flux Calibration Tables”). Ask yourself: **Would a reader struggle to follow or reproduce the results without this information?** If yes → it belongs in the main **Methods** section, not in an appendix.

Master vs. PhD thesis*

Master thesis:

- A demonstration that you can apply scientific methods.
- 1 to 2 years
- ~50 pages or ~20 000 words from Intro to Bibliography.
- One science project (IMRED)

PhD thesis:

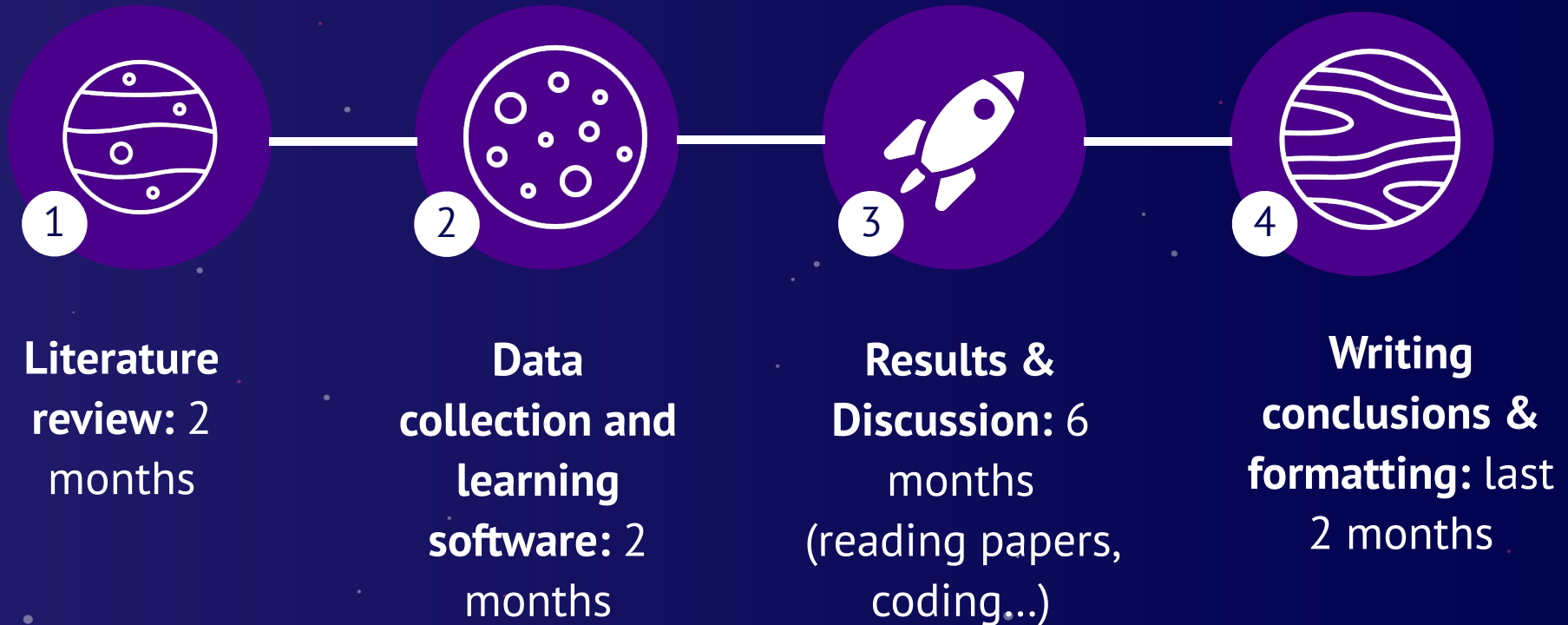
- A demonstration that you can produce original knowledge.
- 3-5 years
- ~150 - 250 pages or ~80 000 words.
- 2 to 4 science projects.
- Monograph thesis (one IMRED)
- Thesis by publication or paper based
 - Each project into IMRED
 - One overall Intro/lit. Review
 - One overall summary and conclusion.

* Again, please refer to your university requirements for more detailed requirements for formatting (e.g., number of pages, page margins, etc.)

Timeline and planning

No fixed schedule, you have to manage your own time!!!!

12 Months



Be flexible, do not push all your writings into the end!!! Set **realistic goals**.
If you finish something, take some notes or write about it immediately

Practical Research Habits

Some tips I wish I had known earlier

- Read every day to stay up-to-date (ArXiv is your friend)
- Write weekly reports (what did you do), be it for your supervisor or just for you to reflect.
- If you finish something, **WRITE** about it immediately! Otherwise you **WILL** forget the details.
- Track your progress weekly, monthly: set **GOALS** and make them visible.
- Try breaking the goals into small steps as possible
- WRITE down every single step you take during coding or analysis. These notes might become your appendices and you might even create teaching materials later.



Practical Research Habits

Monthly Reflections



- Do monthly recap every month
- What came easiest this month? Why?
- What felt hardest?
- Recognize when do you work best? (At a cafe, at night, in the morning...)

Community Writing Practices

Some suggestions:

“Writing can be lonely. Why not write together?”

- A monthly 1h30 writing session with peers. No distractions.
- A monthly feedback session where you share a piece of writing or a figure and get comments.



Some useful tools

For managing references:

- Zotero
- Mendeley

For taking notes, manage tasks

- Notion
- OneNote

For tracking how you spend time

- Clockify
- Focus Plant

For project management

- Trello
- Gantt charts

For writing

- Overleaf
- Or just use LaTeX editors.



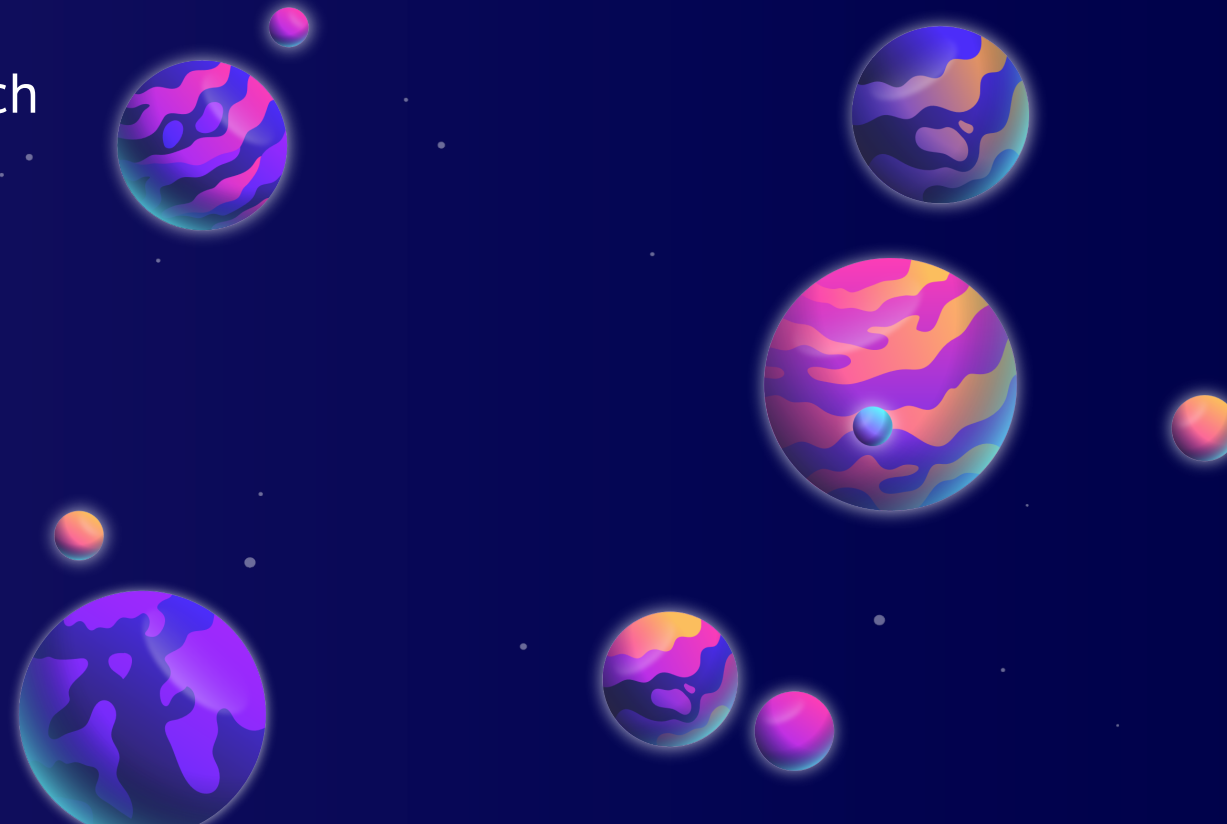
Mini-Activity (15 mn)

Know your project and/or research interests

- ❖ You need to know your science goals (what are you going to contribute to the community)
- ❖ You need to know the gaps/limitations in your field

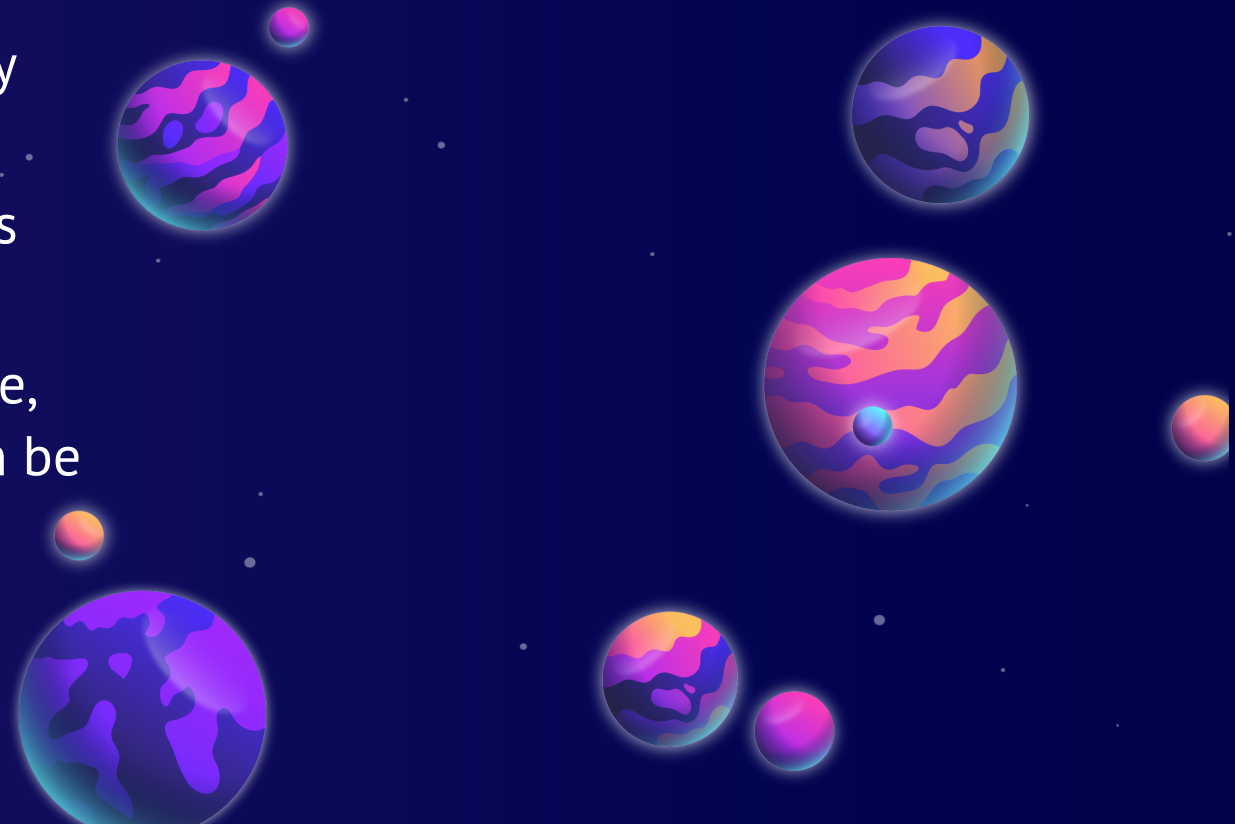
Take 5 mn and write:

- If you don't yet have a topic: What are your research interests?
- If you do: What is your topic, and what is your scientific contribution?



Final notes

- ❖ Be patient with yourself.
- ❖ Be willing to learn new things.
- ❖ It's okay not to know things yet.
- ❖ Most of the tools I use today I learned during my PhD or postdoc. It is never too late!
- ❖ Stay curious. Share with peers. And ask questions often.
- ❖ If you are stuck somewhere, taking a break is fine, but do not disappear! I know doing research can be overwhelming. Ask around, tell your supervisor.





**You've got
this!!!!**



Questions and/or CV feedback

Some useful resources

- [How to write a thesis in LaTeX](#)
- <https://www.compilatio.net/en/blog/thesis>
- [Cool tips for thesis writing](#)
- [An Insider's guide on how to write a thesis when you're short on time](#)

